W-C-440B
August 11, 1970
SUPERSEDING
Fed. Spec. W-C-440A
February 5, 1964

FEDERAL SPECIFICATION

CLIPS, ELECTRICAL, GENERAL SPECIFICATION FOR

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers electrical clips, insulated and uninsulated.

1.2 Classification.

1 2.1 Types. Electrical clips shall be of the following types

Type PC	-	Power clip
Type PCL	-	Power clip with lug
Type NC	-	Needle clip
Type TCM	-	Test clip, miniature
Type CC	-	Crocodile clip
Type TCI	-	Test clip, insulated
Type TC	-	Test clip
Type TCC	-	Test clip, clamping ears
Type BC	-	Battery clip

2. APPLICABLE DOCUMENTS

2.1 Specifications and Standards. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FEDERAL SPECIFICATIONS

QQ-C-530	-	Copper-Beryllium Alloy Bar, Rod, and Wire (Copper Alloy Number 172).
QQ-W-321	-	Wire, Copper Alloy
QQ-W-405	-	Wire, Steel, Carbon and Alloy (for Cold Heading and Cold Forging).
QQ-W-428	-	Wire, Steel, High Carbon, Round, Uncoated for Mechanical Springs (General Purpose).
QQ-W-470	-	Wire, Steel, Carbon, Spring, Music.
ZZ-R-765	-	Rubber, Silicone, Low- and High- Temperature and Tear Resistant.
PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-636	-	Box, Fiberboard.
PPP-B-665	-	Boxes, Paperboard, Metal Stayed (Including Stay Material)
PPP-B-676	-	Boxes, Set-up, Paperboard.
PPP-T-60	-	Tape, Pressure-Sensitive Adhesive, Waterproof, for Packaging.
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

FEDERAL STANDARDS

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications and Standards, and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

MILITARY SPECIFICATIONS

MIL-P-116 - Preservation, Methods of.
MIL-C-45662 - Calibration Systems Requirements.

MILITARY STANDARDS

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification, Marking of U. S. Military Property.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards Handbook:

H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Material. Material shall be as specified herein. When a definite material is not specified, a material shall be used which will enable the electrical clips to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.1.1 Insulation.

3.1.1.1 Rubber. Rubber shall conform to class III, grade 50 of ZZ-R-765 and shall not contain reclaimed rubber.

3.1.1. Polyvinyl chloride. Polyvinyl material shall be opaque nonrigid vinyl chloride-acetate having a durometer hardness not over 60 shore type "A" test and the dielectric strength shall be 250 volts per mil of thickness minimum. After aging in oil for 8 hours at 100° ±1.1°C (212° ±2°F), polyvinyl material shall not be tacky nor show signs of decomposition, and when flexed, shall not crack nor split

3.1.2 Metals.

3.1.2.1 Brass Brass shall conform to QQ-W-321.

3.1.2.2 Phosphor bronze. Phosphor bronze shall conform to QQ-W-321

3.1.2.3 Copper. Copper material as specified herein shall be commercially pure copper, or copper alloyed with 3 to 5 percent zinc.

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3.1.2.4 Copper-beryllium alloy. Copper-beryllium alloy shall conform to QQ-C-530.

3.1.2.5 Steel. Steel shall conform to QQ-W-405, QQ-W-428, or QQ-W-470

3.2 Design and construction. Electrical clips shall be of the design, construction, and physical dimensions specified (see 3.7).

3.2.1 Insulator sets. When specified (see 6.1), clips shall be furnished with an insulator set molded of a polyvinyl or silicone rubber material having such dimensions that will provide a suitable fit. When required, insulator sets shall be installed. Unless otherwise specified (see 6.1), these insulator sets shall be furnished half red and half black for polarity identification purposes

3.2.2 Threaded parts. All threaded parts shall be in accordance with National Bureau of Standards Handbook H28.

3.2.2 1 Engagement of threaded parts. Threaded parts shall engage by at least two full threads.

3.2.3 Attaching hardware. Unless otherwise specified, machine screws and rivets shall be fabricated from carbon steel in accordance with QQ-W-405 except Bessemer compositions (55,000 lb_f/in^2 minimum ultimate strength) Brass screws shall be fabricated in accordance with QQ-W-321.

3 2.4 Finish. Electrical clips shall be unplated, except that ferrous materials shall be suitably plated to resist corrosion.

3.2.5 Spring. The spring or clip shall be designed so that the spring will not become dislodged when clip is fully compressed.

3.2.6 Minimum metal thickness Copper parts of electrical clips shall be of minimum thickness as specified in detail requirements for individual clip types.

3.2.7 Jaws. There shall be free jaw openings. In designs where jaws mesh, all teeth shall enter in the mating trough In designs where jaws pass, the front turn-over portions of the jaws shall not cause binding, and the maximum space between the end of the outer jaws turn-over portion and the inner jaw turn-over portion shall not exceed twice the thickness of the metal.

3.2.8 Shunt (where applicable) The shunt shall be placed in such a manner that one coil of the spring shall rest on each end of the shunt so that the shunt will not become dislodged.

3.2.9 Wire crimping ears. The wire crimping ears shall be of sufficient size that they will accommodate a wire capable of carrying the maximum current for which the clip is intended.

3.2.10 Hinge pins Only solid rivets shall be used for hinge pins.

3.3 Spring pressure. When electrical clips are tested as specified in 4.6.2, the minimum opening force required shall be as specified on the applicable figure.

3.4 Salt-spray (corrosion). When electrical clips with attaching hardware are tested as specified in 4.6.3, there shall be no exposure of the plated base metals or blistering of plated surfaces of the hardware.

3.5 Identification marking. Electrical clips shall be marked in accordance with MIL-STD-130 with the type designation and item number, if applicable, the manufacturer's name, trademark, or code, and, if space permits, the current rating (amps)

3.6 Workmanship

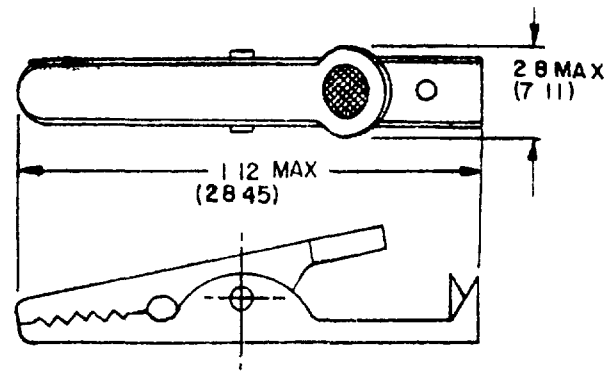
3.6.1 Electrical clips Electrical clips shall be free from burrs, sharp edges, flaws, fractures or cracks in the metal. The clips shall be free of grease oil, metal filings, chips and other foreign material. They shall be uniform in quality and free from defects that will affect life, serviceability, or appearance.

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3.6.2 Insulator sets. The interior shall be free from mold flash congestion. They shall be uniform in quality and free from defects that will affect life, serviceability, or appearance. Insulator sets shall not restrict free jaw opening of the clips.

3.7 Detail requirements for individual clip types. Detail requirements applicable to individual types of electrical clips shall be as specified on the applicable figure. In the event of any conflict between the general requirements of this specification and the detail requirements, the latter shall govern.

3.7.1 Type TCM electrical clip. Type TCM electrical clips shall be of the design, construction, and physical dimensions specified on figure 1 and shall be furnished with a conductor bridge or hole and insulation clamping ears.



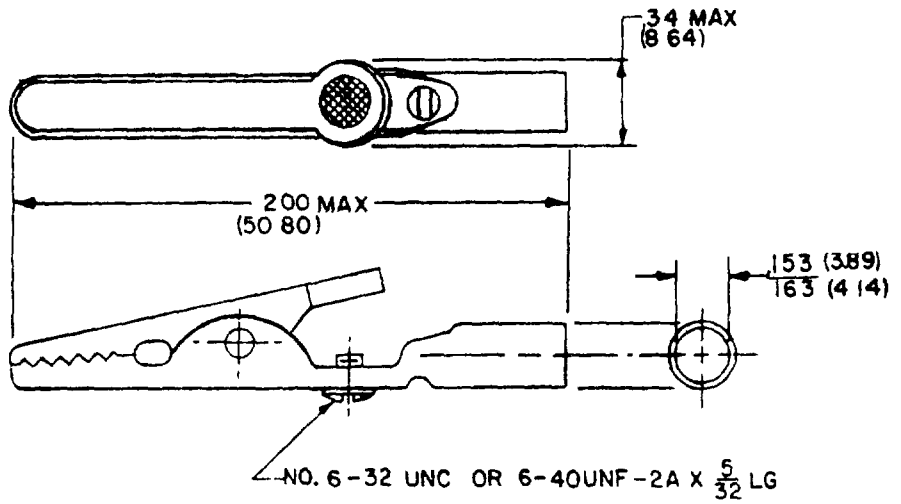
Item No.	Jaw opening (min)	Spring pressure (lbs-min)	Nominal current rating (amp)	Material	Metal thickness (min)
1	.19 (4.83)	2	5	Copper	.0140 (.36)

NOTES:
1. Dimensions are in inches.
2. Drawing depicts only general design characteristics and is not intended to limit specific manufacturing processes.
3. Teeth shall mesh.
4. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
5. Millimeters are in parentheses.

FIGURE 1. Type TCM clip, electrical

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3.7.2 Type TC electrical clip. Type TC electrical clips shall be of the design, construction, and physical dimensions specified on figure 2 and shall be furnished with a screw-type wire accommodation, conductor bridge, and a banana plug facility.



Item No.	Jaw opening (min)	Spring pressure (lbs-min)	Nominal current rating (amp)	Material	Metal thickness (min)
1	.31 (7.87)	3½	10	Copper	.0186 (.47)

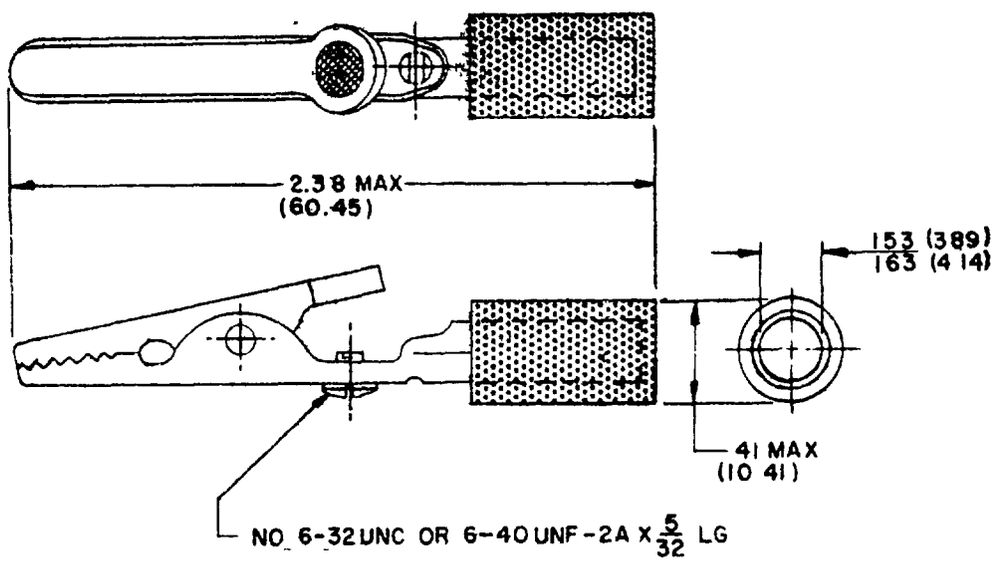
NOTES
1. Dimensions are in inches.
2. Drawing depicts only general design characteristics and is not intended to limit specific manufacturing processes.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm
4. Millimeters are in parentheses
5. Teeth shall mesh.

FIGURE 2 Type TC clip, electrical

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3.7.3 TCI electrical clip. Type TCI electrical clips shall be of the design, construction, and physical dimensions specified on figure 3 and shall be furnished with a screw-type wire accommodation, conductor bridge, and banana plug facility.

3.7.3.1 Insulated portions. Insulated portions shall be cellulose acetate or nylon.



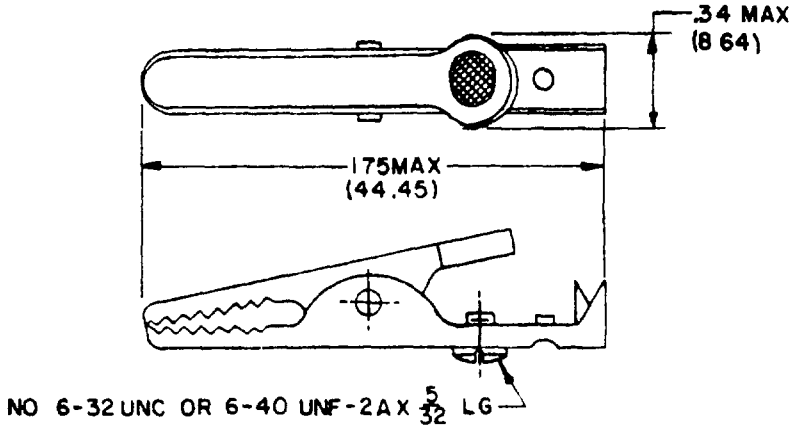
Item No.	Jaw opening (min)	Spring pressure (lbs-min)	Nominal current rating (amp)	Insulator color	Material	Metal thickness (min)
1	.31 (7.87)	3½	10	Red	Copper	.0186 (.47)
2				Black		

NOTES
1. Dimensions are in inches.
2. Drawing depicts only general design characteristics and is not intended to limit specific manufacturing processes
3. Shaded area indicates insulated portion.
4. Teeth shall mesh.
5. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm
6. Millimeters are in parentheses.

FIGURE 3. Type TCI clip, electrical

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3.7.4 Type TCC electrical clip. Type TCC electrical clips shall be of the design, construction, and physical dimensions specified on figure 4 and shall be furnished with a screw-type wire accommodation, conductor bridge, and insulation clamping ears.



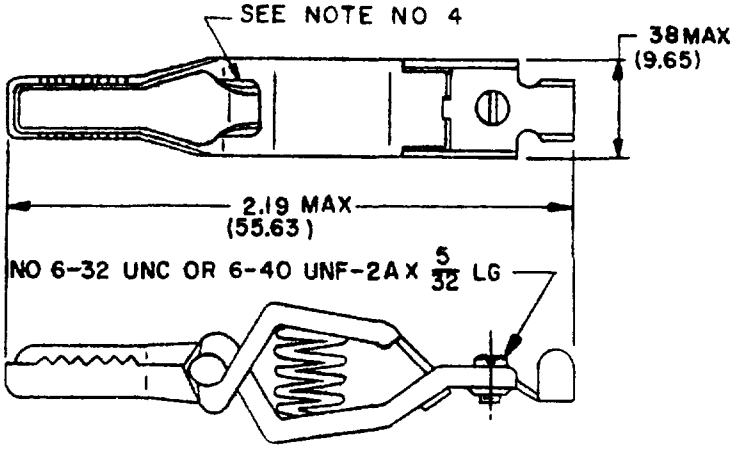
Item No.	Jaw opening (min)	Spring pressure (lbs-min)	Nominal current rating (amp)	Material	Metal thickness (min)
1	.31 (7.87)	3	10	Copper	.0186 (47)

NOTES
1. Dimensions are in inches.
2. Drawing depicts only general design characteristics and is not intended to limit specific manufacturing processes
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm
4. Millimeters are in parentheses.

FIGURE 4 Type TCC clip, electrical

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3.7.5 Type CC electrical clip. Type CC electrical clips shall be of the design, construction, and physical dimensions specified on figure 5 and shall be furnished with a screw-type wire accommodation and insulation clamping ears.



Item No.	Jaw opening (min)	Spring pressure (lbs-min)	Nominal current rating (amp)	Material	Metal thickness (min)
1	.31 (7.87)	4½	10	Copper	.0260 (.66)

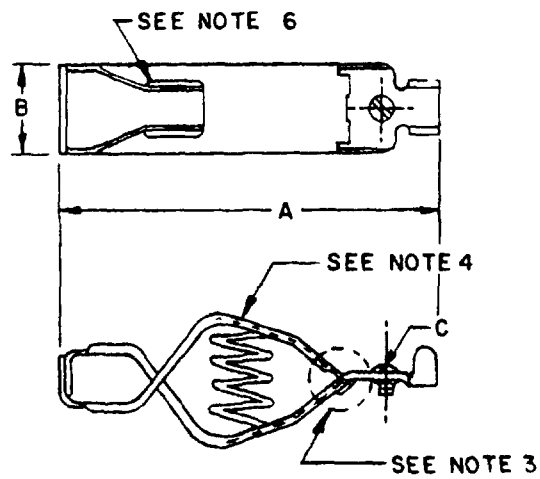
NOTES.
1. Dimensions are in inches.
2. Drawing depicts only general design characteristics and is not intended to limit specific manufacturing processes.
3. Item No. 1 shall have a phosphor bronze spring and a brass screw.
4. Interlocking neck construction is required.
5. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
6. Millimeters are in parentheses.

FIGURE 5. Type CC clip, electrical

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3.7.6 Type PC electrical clip. Type PC electrical clips shall be of the design, construction, and physical dimensions specified on figure 6 and shall be furnished with a screw-type conductor accommodation for wire.

3.7.6.1 Hinge life (PC clips only). Hinge shall be constructed in such a manner that no damage will result nor will hinge become disassembled when subjected to test of paragraph 4.6.4.



Item No.	A (max)	B (max)	C Attaching hardware	Jaw opening (min)	Spring pressure (lbs) (min)	Nominal current rating (amp)	Material	Metal thickness (min)
1	1.62' (41.15)	.31 (7.87)	6-32 UNC or 6-40 UNF 2A x 5/32 lg	.25 (6.35)	4	10	Copper	.0238 (60)
2	2.12 (53.85)	.38 (9.65)	6-32 UNC-2A x 3/16 lg.	.44 (11.18)	4	20	Copper	.0260 (.66)
3	2.44 (61.98)	.50 (12.70)	6-32 UNC-2A x 3/16 lg.	.62 (15.75)	7	40	Copper	.0310 (.79)
4	3.12 (79.25)	.69 (17.53)	8-32 UNC-2A x 1/4 lg.	.75 (19.05)	9	50	Copper	.0345 (.88)

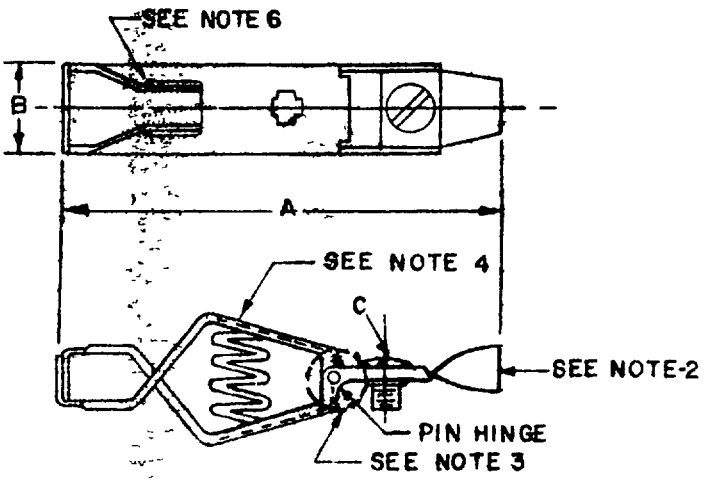
NOTES.
1. Dimensions are in inches.
2. All items have crimp ears.
3. Design of pivot is optional.
4. Item No. 4 shall incorporate a solid copper shunt positioned to prevent overheating of the spring
5. Drawing depicts only general design characteristics and is not intended to limit specific manufacturing processes.
6. Interlocking neck construction is required.
7. Item No. 1 shall have a phosphor bronze spring and a brass screw
8. Insulator sets (see 3.2.1)
9. Millimeters are in parentheses
10. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 6. Type PC clip, electrical

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3.7.7 Type PCL electrical clip. Type PCL electrical clips shall be of the design, construction, and physical dimensions specified on figure 7 and shall be furnished with a screw-type conductor accommodation for wire or terminal lug (see table).

3.7.7.1 Hinge pin. The upset end shall be concentric with the rivet diameter and shall have like appearance to the rivet head.



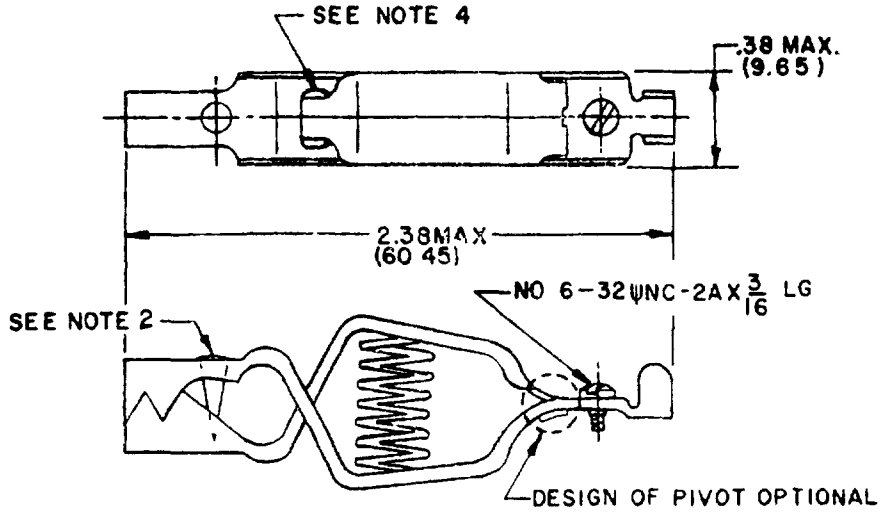
Item No	A (max)	B (max)	C Attaching hardware	Jaw opening (min)	Spring pressure (lbs) (min)	Nominal current rating (amp)	Material	Metal thickness (min)	Lugs		
									Inside dia	AWG cable size (min)	Wall thkns (min)
1	4.75 (120.65)	.81 (20.57)	10-32UNF-2A 5/16 lg	1.06 (26 92)	18	100	Copper	0478 (1.21)	353 (8.97)	No 4	.0420 (1.07)
2	6.25 (158.75)	1.12 (28.45)	1/4 20UNC 2A 7/16 lg , washer lock, split for size 1/4 screw nut, hex 1/4-20UNC-2B	1.62 (41.15)	24	200	Copper	.0610 (1.55)	545 (13.84)	No 1	0420 (1.07)
3	8 00 (203.20)	1.44 (36.58)	5/16-18UNC-2A x ½ lg., washer, lock split for size 5/16 screw nut, hex 5/16-18UNC-2B	2.00 (50.80)	35	300	Copper	0750 (1.91)	668 (16.97)	No 4/0	.0720 (1.83)

NOTES
1. Dimensions are in inches.
2. Items have solder type, copper lugs.
3. A pin hinge is required.
4. All items shall incorporate a solid copper shunt positioned to prevent overheating of the spring.
5. Drawing depicts only general design characteristics and is not intended to limit specific manufacturing processes.
6. Interlocking neck construction is required.
7. Insulator sets (see 3.2.1).
8. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch 25.4 mm.
9. Millimeters are in parentheses

FIGURE 7. Type PCL clip, electrical

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3.7.8 Type NC electrical clip. Type NC electrical clips shall be of the design, construction, and physical dimensions specified on figure 8 and shall be furnished with a screw-type conductor accommodation and insulation clamping ears.



Item No.	Jaw opening (min)	Spring pressure (lbs-min)	Nominal current rating (amp)	Material	Metal thickness (min)
1	.50 (12.70)	6	20	Copper	.0300 (.76)

NOTES
1. Dimensions are in inches.
2. The insulation piercing pin shall be copper-beryllium alloy ground to a sharp point
3. Drawing depicts only general design characteristics and is not intended to limit specific manufacturing processes.
4. Interlocking neck construction is required.
5. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch 25.4 mm.
6. Millimeters are in parentheses

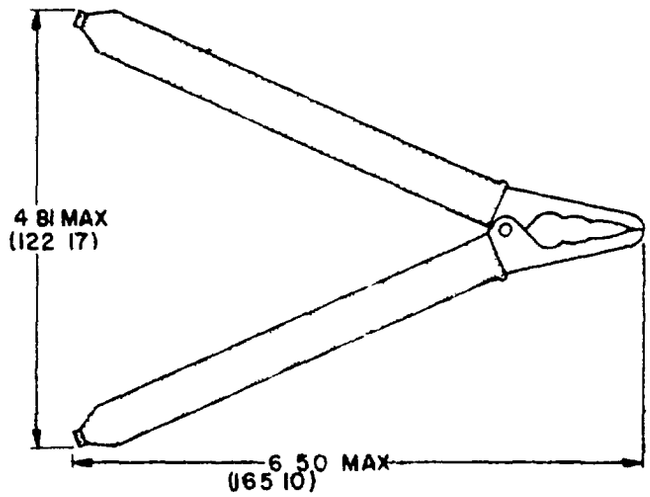
FIGURE 8 Type NC clip, electrical

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3.7.9 Type BC electrical clip. Type BC electrical clips shall be of the design, construction, and physical dimensions specified on figure 9 and shall be furnished with both screw-mounted, solder-type lug and an integral solder bridge conductor accommodation. Insulating clamping ears shall also be provided.

3.7.9.1 Insulated portions. Insulated portions shall be nonrigid polyvinyl-chloride.

3.7.9.2 Hinge pin. The upset end shall be concentric with the rivet diameter and shall have like appearance to the rivet head.



Item No.	Jaw opening (min)	Spring pressure (lbs-min)	Nominal current rating (amp)	Insulator color	Material	Metal thickness (min)
1	1.50 (38.10)	10	200	Red	Copper	.0480 (1.22)
2				Black		

NOTES
1. Dimensions are in inches.
2. Shaded areas indicates insulated portion.
3. Drawing depicts only general design characteristics and is not intended to limit specific manufacturing processes.
4. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
5. Millimeters are in parentheses

FIGURE 9. Type BC clip, electrical

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the supplier. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) Materials inspection (see 4.3).
(b) Quality conformance inspection (see 4.5).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table II, used in fabricating the electrical clips, are in accordance with the applicable referenced specifications or requirements prior to such fabrication

TABLE II. Materials inspection

Material	Requirement paragraph	Applicable specification
Rubber	3.1.1.1	ZZ-R-765
Polyvinyl	3.1.1.2	
Metals:		
Brass	3.1.2.1	QQ-W-321
Phosphor bronze	3.1.2.2	QQ-W-321
Copper	3 1.2.3	
Copper-beryllium alloy	3.1 2.4	QQ-C-530
Steel screw, attaching hardware and rivets	3.2.3	QQ-W-405
Rivets	3.2.3	QQ-W-405
Spring wire steel	3.1.2.5	QQ-W-405, QQ-W-428, or QQ-W-470
Brass screws	3.2.3	QQ-W-321

4.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202

4.5 Quality conformance inspection.

4 5.1 Inspection of products for delivery. Inspection of products for delivery shall consist of groups A and B inspection Except as specified in 4 5.2.1.4, delivery of products which have passed the groups A and B inspections shall not be delayed pending the results of the group C inspection.

5.1.1 Inspection lot. An inspection lot shall consist of all the electrical clips of the same type size produced under essentially the same conditions and offered for inspection at one time.

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4.5.1.2 Group A inspection. Group A inspection shall consist of the examinations specified in table III.

4.5.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level I. The acceptable quality level (AQL) shall be as specified in table III. Major and minor defects shall be as defined in MIL-STD-105.

4.5.1.2.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

TABLE III. Group A inspection

Examination	Requirement paragraph	Test method paragraph	AQL (percent defective)	
			Major	Minor
Visual and mechanical examination	3.1 to 3.2.10 incl, 3.5 and 3.6	4.6.1	1.0	4.0

4.5.1.3 Group B inspection. Group B inspection shall consist of the tests specified in table IV, and the sample shall be selected from inspection lots that have passed group A inspection.

4.5.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105 for general inspection level I. The sample size shall be based on the inspection lot size from which the sample was selected for group A inspection. The AQL shall be 4.0 percent defective.

4.5.1.3.2 Rejected lots If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

TABLE IV. Group B inspection

Test	Requirement paragraph	Test method paragraph
Spring pressure Hinge life (PC clips only)	3.3 3.7.6.1	4.6.2 4.6.4

4.5.1.3.3 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract or order.

4.5.2 Periodic inspection. Periodic inspection shall consist of group C inspection.

4.5.2.1 Group C inspection. Group C inspection shall consist of the salt spray (corrosion) (see 3 4) test as specified in 4.6.3. Group C inspection shall be made on sample units selected from inspection lots which have passed the groups A and B inspection.

4.5 2.1.1 Sampling plan. Six sample units shall be selected from those covered by a single type and size. Inspection shall be made on the first lot and once each 12 months thereafter.

4.5.2.1.2 Failures. If one or more sample units fail to pass group C inspection, the sample shall be considered to have failed.

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4.5.2.1.3 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract or purchase order.

4.5 2.1.4 Noncompliance. If a sample fails to pass group C inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Groups A and B inspections may be reinstituted; however, final acceptance shall be withheld until the group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.5.3 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation and packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

4 6 Method of examination and test.

4.6.1 Visual and mechanical examination. Electrical clips and insulator sets shall be examined to verify that the design, construction, physical dimensions, and markings are in accordance with the applicable requirements. (See 3.1 to 3.2.10 incl, 3.5 and 3.6)

4.6.2 Spring pressure (see 3.3). Electrical clips shall be tested for spring pressure in accordance with the following details:

(a) Mounting. The electrical clips shall be mounted in a fixture so designed as to permit accurate measurement of the jaw opening and pressure required to open the jaws to one-half the specified minimum jaw opening.
(b) Procedure. The electrical clips shall be opened to the specified minimum jaw opening three times, after which the jaws shall be opened to one-half the minimum opening three times with measurements being taken each time to determine whether the minimum jaw opening and the spring pressure required to open the jaws to one-half the minimum are as specified on the applicable figure (see 3.3). Specified spring pressure for types TCM, TC, TCI, and TCC shall be applied at the farthermost point on the thumb pad from the hinge pin. For types CC, PC, PCL, and NC, pressure shall be applied exactly over the center axis of the helical springs. For BC type, pressure shall be applied at the tips of the handles.

4.6.3 Salt spray (corrosion) (see 3.4). Electrical clips containing ferrous material shall be tested in accordance with method 101 of MIL-STD-202. The following details shall apply:

(a) Test-condition letter B.
(b) Measurements after exposure. Ferrous parts shall be visually examined for exposure of plated base metal or blistering of plated surfaces.

4.6 4 Hinge life (PC clips only) (see 3.7.6.1) The clips shall be clamped to a steel mandrel of a diameter equal to the minimum jaw opening specified in figure 6. Remove the clips and allow the jaws to come to rest. The above procedure shall be repeated for 100 operations. The axis of the clip and mandrel shall be in a straight line. At the end of the test, the clip shall be examined to determine that there is no loosening of parts and requirements of paragraph 3.7.6.1 have been met

PREPARATION FOR DELIVERY

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5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning. Electrical clips shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Electrical clips shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packaging. Electrical clips shall be packaged in accordance with MIL-P-116, method III insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein. Unattached hardware, such as insulators, machine screws and rivets shall be physically protected and enclosed within the unit package.

5.1.1.4.1 Quantity per unit package (QUP). Unless otherwise specified (see 6.1), all electrical clips, except types PCL and BC, shall be unit packaged five each. Electrical clips conforming to types PCL and BC shall be unit packaged one each.

5.1.1.5 Intermediate packaging. Electrical clips, packaged as described in 5.1.1.4, shall be placed in intermediate containers conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size, shape and quantities, shall be of minimum tare and cube and shall contain multiples of five unit packages, not to exceed 50 packages. No intermediate packaging is required when the total quantity shipped to a single destination is less than 50 unit packages.

5.1.2 Level B (civil agencies) Ten type PCL, item no. 1 clips shall be packaged in a box conforming to PPP-B-566, PPP-B-665, or PPP-B-676. The box shall be secured to prevent accidental opening.

5.1.3 Level C. Cleaned and dried electrical clips shall be packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. The packaged electrical clips shall be packed in fiberboard containers conforming to PPP-B-636, class weather-resistant, style optional, special requirements. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joint with tape, 2 inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only.

5.2.2 Level B.

5.2.2.1 For civil agencies. Ten unit packages of type PCL, item no. 1 clips (100 clips) or 10 type PCL, item no. 2 clips shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix of PPP-B-636.

5.2.2.2 For military agencies. The packaged electrical clips shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto For Army procurement, fiberboard containers shall be class weather-resistant as specified in level A

5.2.3 Level C. The packaged electrical clips shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

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5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required in the contract or order, each shipping container shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, each unit package and intermediate and exterior container shall be marked in accordance with MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Navy procurements. For Navy procurements the use of polystyrene loose fill material (such as strips, strands and beads) is prohibited for packaging and packing applications.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and item number (see 1.2 and 3.7 to 3.7.9 inclusive)
(c) Whether or not insulator sets are to be furnished with clips (see 3.2.1).
(d) When applicable, the proportion of red and black for insulator sets, if other than half and half (see 3.2.1)
(e) Preparation of insulators or insulator sets when the insulators or sets are ordered separately.
(f) Level of packaging and packing required (see 5.1 and 5.2).
(g) Quantity per unit package (see 5.1.1.4.1)

Custodians
Army - EL
Navy - SH
Air Force - 80

Reviewer
Army - MI
Navy - SH, AS, EC
Air Force - 80
DSA - ES

User
Army -
Navy - OS, CG, MC
Air Force -

Preparing activity
Army - EL

Agent
DSA - ES

(Project 5940-0467)

CIVIL AGENCY INTEREST:

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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